

Week 15

Course Integration & Final Projects

Ten Methods, One Dataset, Three Lessons

Section	Question it answers
1. The Complete Comparison	Which method wins — and when does it matter?
2. The Decision Framework	Given a new dataset, how do you choose?
3. Ethical Considerations	What responsibilities come with clinical ML?

1. The Complete Comparison

Which method wins — and when does it matter?

Over 14 weeks we applied 10 methods to the same reaching dataset: 480 trials, 8 directions, 20 subjects, EMG and neural features. Every method was evaluated with leave-one-subject-out cross-validation on the same train/test splits. Figure 1 displays the results as a heatmap — green is high accuracy, red is low. The patterns are more interesting than any single number.

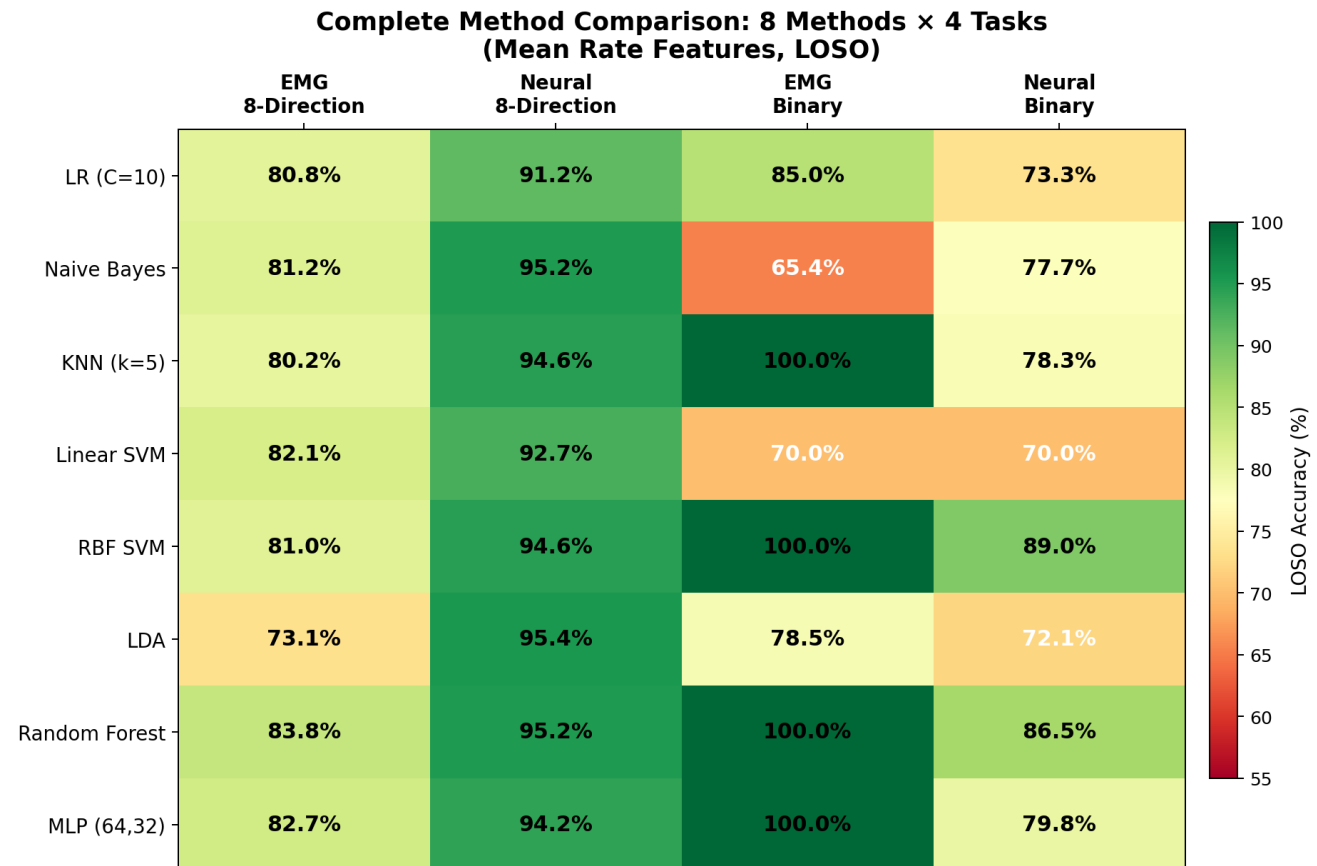


Figure 1. The complete heatmap: 8 methods × 4 tasks (LOSO, mean rate features). Green = high accuracy, red = low. No single method dominates all four tasks.

Three patterns worth discussing

Pattern 1: No single method dominates all four tasks.

RBF SVM and Random Forest achieve 100% on EMG binary diagnosis, but RBF SVM drops to 89.0% on neural binary while RF holds at 86.5%. KNN hits 100% on EMG binary but only 80.2% on EMG 8-direction. LDA leads on neural 8-direction (95.4%) but manages only 78.5% on EMG binary. The best method depends entirely on the task — there is no universal winner.

Pattern 2: Feature set matters as much as the algorithm.

Every method performs better on neural features (80D) than EMG features (6D) for direction decoding: the gap ranges from 10 points (LR) to 14 points (NB). But for binary diagnosis, the pattern reverses for some methods — KNN achieves 100% on EMG but only 78.3% on neural. The 80 neural features contain more information about direction (cosine tuning) but less about impairment (which manifests in synergy structure, not individual neuron rates). Week 14 showed that for neural features — where impairment detection had been the weakest task (70–89%) — the breakthrough came from preserving temporal structure, not from changing the classifier.

Pattern 3: Simple methods are competitive.

LR (Week 5), the first method we learned, achieves 91.2% on neural direction — within 5 points of the best. On EMG binary, it hits 85.0%, outperforming the linear SVM (70.0%) and LDA (78.5%). LR's strength is its simplicity: 640 parameters, interpretable coefficients, fast prediction. For many practical applications, especially in clinical settings where interpretability matters, LR is good enough.

Figure 2 presents the same data as grouped bar charts, making it easier to see which methods cluster together and which stand apart.

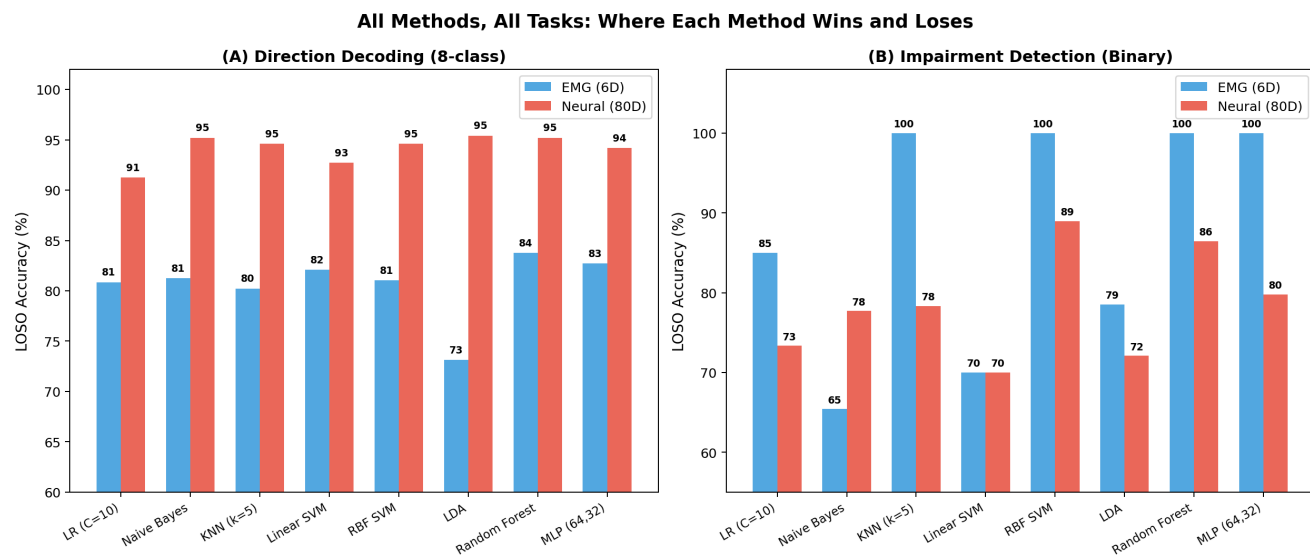


Figure 2. All methods on both tasks. (A) Direction decoding: most methods cluster at 80–83% (EMG) and 91–95% (neural). (B) Binary diagnosis: wide spread from 65% (NB) to 100% (KNN, RBF SVM, RF, MLP) on EMG.

The 10-method comparison table

Figure 3 extends the accuracy heatmap with the properties that matter for choosing a method in practice: parameter count, whether tuning is needed, what interpretability tools are available, and whether the method learns its own features.

Method	Week	Neural 8-Dir	EMG Binary	Params	Tuning?	Interpretability	Learns Features?
NB	7	95.2%	65.4%	~1K	No	Priors	No
LR	5	91.2%	85.0%	640	Yes (C)	Coefficients	No
KNN	8	94.6%	100%	—	Yes (k)	None	No
Lin SVM	9	92.7%	70.0%	~600	Yes (C)	Weights	No
RBF SVM	9	94.6%	100%	~2K	Yes (C,γ)	None	No
LDA	10	95.4%	78.5%	560	No	Loadings	No
RF	11	95.2%	100%	~40K	No	Feature imp.	No
K-means	12	ARI=0.92	N/A	640	Yes (k)	Centroids	No
GMM	12	ARI=0.91	N/A	~1K	Yes (k)	Means+cov	No
MLP	13	94.2%	100%	7,528	Yes	Hidden reps	Yes

Figure 3. The complete 10-method comparison table. Supervised methods report LOSO accuracy on two representative tasks. Unsupervised methods (green) report ARI. The MLP (orange) is the only method that learns its own features.

Computational cost

Accuracy is not the only consideration. A BCI must decode in real time (under 50 ms per prediction). A clinical screening tool used at the bedside must train in seconds, not minutes. Figure 4 compares training and prediction times.

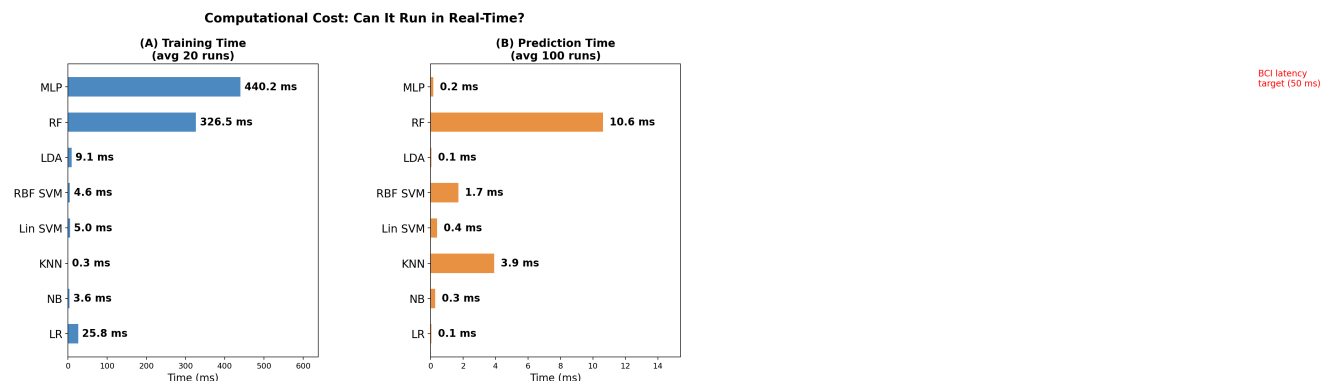


Figure 4. Training and prediction time on the 480-trial dataset. LR and NB predict in under 0.5 ms. RF and MLP are slowest to train (>300 ms) but still fast enough for offline analysis. All methods meet the 50 ms BCI latency target (red line).

At our data scale (480 trials, 80 features), all methods are fast enough for real-time BCI. The differences become critical when scaling to thousands of trials or thousands of features. RF training time scales linearly with trees and features; MLP training scales with epochs and architecture; KNN prediction scales with training set size. LR and LDA remain fast regardless of scale — another reason to try them first.

What each method taught us

Beyond accuracy, each method illuminated a different aspect of the data:

- PCA (Week 4): Healthy subjects need 3 synergies for >90% VAF; impaired need only 2. Dimensionality reduction as scientific discovery.
- LR (Week 5): Established the baseline (80.8% EMG, 91.2% neural). Coefficients reveal which muscles/neurons drive the classification.
- GLM (Week 6): Poisson encoding models recovered cosine tuning curves. Population vectors decoded direction without any ML.
- NB (Week 7): The independence assumption fails on correlated muscles (65.4% EMG binary) but PCA decorrelation rescues it (85%). Assumptions matter.
- KNN (Week 8): 100% on EMG binary — the healthy/impaired clusters are perfectly separated in 6D. But KNN collapses in high dimensions.
- SVM (Week 9): The margin principle: wider gap = more robustness to electrode drift. Linear SVM is the most drift-robust linear method.
- LDA (Week 10): Supervised projection that separates directions better than PCA. LDA loadings show which neurons contribute most.
- RF (Week 11): Feature importance identified the most diagnostic muscles and neurons. 100% on EMG binary with built-in interpretability.
- K-means/GMM (Week 12): Unsupervised clustering recovered 8 directions (k-means ARI=0.92, GMM ARI=0.91) without labels. The ‘subtypes’ illusion: clusters in impaired data were directions, not subtypes.
- MLP (Week 13): Hidden representations separate directions more cleanly (silhouette 0.621) than PCA or LDA, but accuracy is similar (~94%). On 480 trials, learned features do not outperform hand-crafted ones.
- Temporal features (Week 14): LR on 4,000 temporal features (99.0%) beat MLP on 80 mean rates (79.8%) by 19 points. Features matter more than the classifier.

2. The Decision Framework

Given a new dataset, how do you choose?

The first question is not “which classifier?” but “what features?” Week 14’s central result — LR on temporal features (99.0%) beating RF on mean rates (86.5%) by over 12 points — proves that feature design dominates classifier choice. Before reaching for a complex model, ask whether richer features would solve the problem with a simpler one.

Figure 5 distils the decision process into a flowchart. The path depends on four questions: do you have labels? what kind of features? how much data? do you need interpretability?

The Decision Framework: Choosing the Right Method

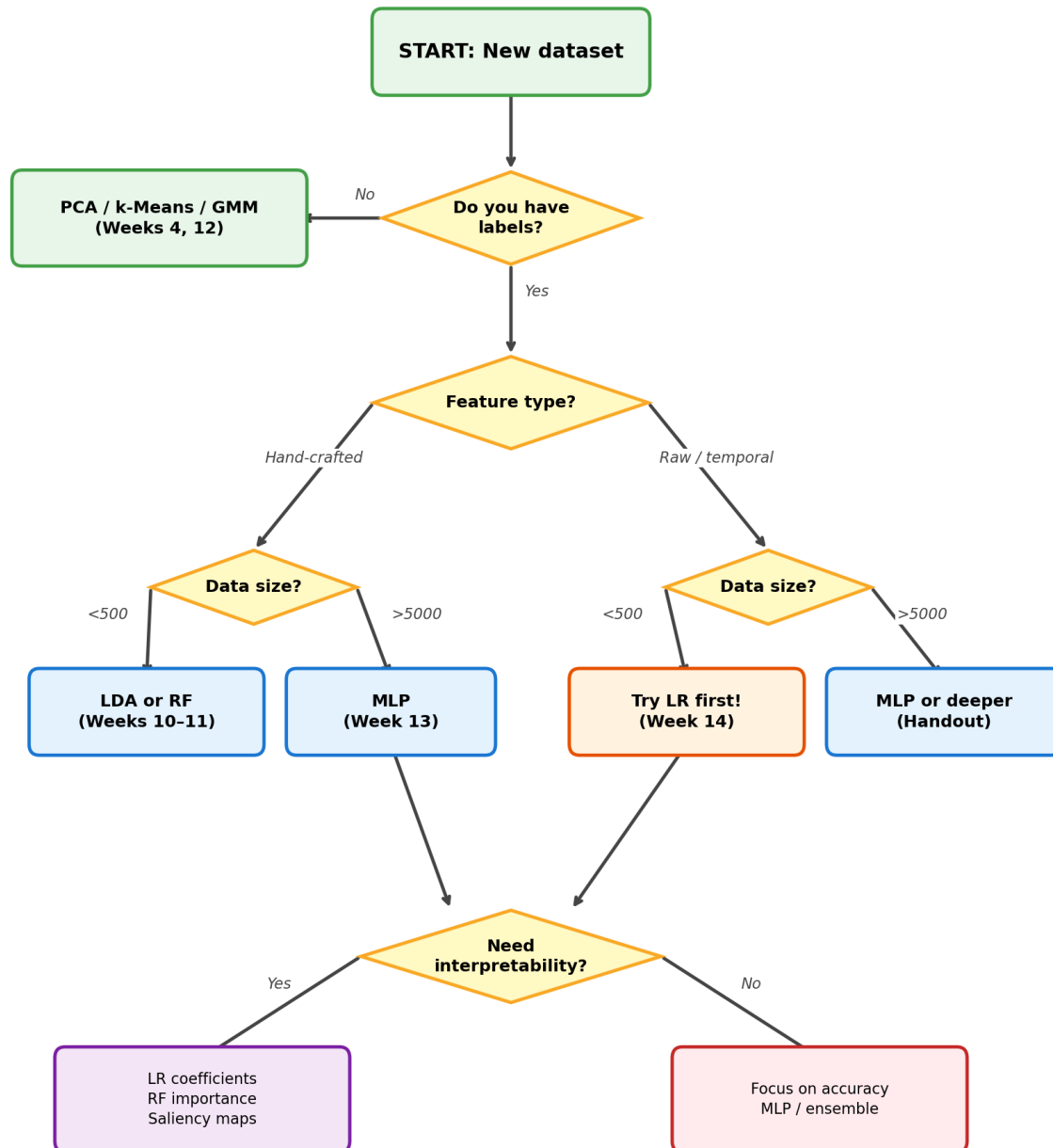


Figure 5. The decision framework. Start with the data: labels? feature type? data size? Each path leads to a recommended method, with interpretability as a secondary consideration.

Walking through the flowchart

No labels → PCA, k-means, or GMM (Weeks 4, 12)

You cannot use supervised methods without ground truth. PCA finds the axes of maximum variance (Week 4); k-means and GMM discover clusters (Week 12). These are exploratory tools — use them when you do not know what structure the data contains.

Hand-crafted features, small data (<500 trials) → LDA or RF

When you have designed good features (mean rates, synergy scores, kinematic variables), classical methods work as well as neural networks. LDA provides interpretable projections; RF provides feature importance rankings. Both matched or beat the MLP on our 480-trial dataset (Weeks 10–11, confirmed in Week 13).

Hand-crafted features, large data (>5,000 trials) → MLP

With enough data, the MLP's learned features begin to outperform hand-crafted ones because it can discover nonlinear interactions that PCA and LDA cannot. Week 13 showed the MLP's hidden representations were already richer (silhouette 0.621 vs 0.456) on 480 trials — with 5,000+ trials, this advantage translates to accuracy gains.

Raw/temporal features, small data → try LR first

This is Week 14's surprising lesson. When the features are high-dimensional but the signal is linearly separable (as our onset-delay impairment signal was in 4,000D), LR with its one-weight-per-feature approach can outperform the MLP. Try LR first; if its linear assumption fails, move to MLP.

Raw features, large data (>5,000 trials) → MLP or deeper

This is where deep learning earns its keep: RNNs for temporal sequences, CNNs for spatial patterns, transformers for population-level dynamics. The accompanying handout 'Beyond This Course' covers these architectures with code examples and references for independent study.

Need interpretability? → LR coefficients, RF importance, saliency maps

A clinician needs to explain why a patient was classified as impaired. LR coefficients point to specific muscles or time bins. RF importance ranks features by predictive value. MLP saliency maps (Week 14) show which neuron-time combinations the network attends to. All three are available regardless of which classifier you choose for the final prediction.

Need drift robustness? → depends on feature type

On mean rates, the MLP's distributed representations provide natural tolerance (Week 13, Figure 13). On temporal features, LR is actually the most robust (Week 14, Figure 11) because 4,000 individual weights provide redundancy. The linear SVM's margin provides a geometric robustness buffer (Week 9, Figure 10). There is no single 'most robust' method — it depends on the feature space.

3. Ethical Considerations

What responsibilities come with clinical ML?

The methods from this course are powerful enough to classify patients, decode intended movements, and predict clinical outcomes. That power comes with responsibilities that go beyond cross-validation accuracy.

Data privacy in clinical neuroscience

Neural recordings are among the most personal data imaginable. A motor cortex recording captures not just intended movements but also cognitive states, attention fluctuations, and potentially emotional valence. Key considerations:

- De-identification is harder than it appears. Even after removing names and dates, neural firing patterns may be individually identifiable — like a neural fingerprint. With 80 neurons and 480 trials, a well-trained classifier might identify which subject produced a given trial with high accuracy, re-identifying ‘anonymous’ data.
- Informed consent must cover ML uses. A subject who consents to ‘a motor control study’ may not expect their data to train a classifier that could later be used for clinical diagnosis. Consent forms should specify the range of analyses planned, including machine learning applications.
- Data sharing requires safeguards. Open-science norms encourage sharing datasets, but neural data requires more careful handling than behavioural data. Differential privacy, data use agreements, and access-controlled repositories are minimum standards.

Algorithmic bias in diagnostic tools

Our dataset contains 10 healthy and 10 impaired subjects. In a real clinical setting, the population is far more diverse, and several biases can creep in:

- Sample bias. If the training data over-represents one demographic (e.g., young male subjects from a university population), the classifier may perform poorly on older patients, women, or people with different body compositions — exactly the populations most likely to need clinical assessment.
- Label bias. The ‘impaired’ label in our dataset is binary, but real impairment exists on a spectrum. A classifier trained on severe impairment may miss mild cases, creating a false sense of confidence in screening accuracy.
- Feature bias. EMG features depend on electrode placement, skin properties, and subcutaneous fat — all of which vary systematically across demographics. A classifier that learns to associate high impedance with impairment is actually learning a confound, not a diagnostic signal.
- Evaluation bias. LOSO cross-validation protects against within-subject data leakage, but it does not protect against demographic confounds. If all impaired subjects happen to be older, the classifier may be decoding age, not impairment. Stratified evaluation across demographic groups is essential.

Responsible reporting of ML results

The neuroscience literature increasingly includes ML results, but reporting standards vary widely. Best practices:

- Report the full pipeline, not just accuracy. Preprocessing steps (standardisation, PCA), hyperparameter selection strategy (grid search, cross-validation), and the exact train/test split protocol (LOSO, k-fold) must all be reported. A 95% accuracy with trial-level k-fold (data leakage) is not comparable to 95% with LOSO.
- Report confidence intervals or variability. A single LOSO accuracy number hides the variance across folds. Some subjects may be classified perfectly while others are at chance. Per-subject accuracy distributions are far more informative.
- Compare against meaningful baselines. Claiming 95% accuracy is meaningless without context. Report chance level (12.5% for 8 directions, 50% for binary), the simplest reasonable baseline (LR), and existing state-of-the-art if applicable.
- Distinguish prediction from explanation. A random forest that achieves 100% on EMG binary does not prove that its top-ranked features are causal. Feature importance is a description of what the model uses, not a mechanism. Causal claims require experimental manipulation, not classification accuracy.
- Be honest about limitations. Our MLP matched but did not beat classical methods. That is a result worth reporting — it saves other researchers from assuming neural networks are always better. Negative results and null differences are valuable.

The broader context

As ML tools become standard in neuroscience, the field must develop norms around reproducibility, fairness, and transparency. Several initiatives are underway: the FAIR principles for data sharing, model cards for documenting ML systems, and pre-registration of analysis plans to prevent p-hacking via method shopping. Students leaving this course should be not only technically competent but also ethically informed — capable of building classifiers that work and knowing when it is appropriate to deploy them.

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